



ALLIANCE SCHOOL OF ADVANCED COMPUTING

Departments of CSE, CSE Specialisations and IT

DESIGN PROJECT — GUIDELINES AND EVALUATION RUBRICS

Design Project – 1 (5CS1990 / 5IT1990, Semester V) & Design Project – 2 (6CS1991/ 6IT1991, Semester VI)

Design Project – 1 (E1CSA313 / E1ITA306) & Design Project – II ()

Guidelines

1. Projects must be approved by the Mentor before starting of the process.
2. All works must be original; plagiarism is strictly prohibited.
3. Students can work in teams.
4. Teams must have 4 members.
5. Roles and contributions of each team member must be clearly defined.
6. A team leader must be designated for communication and coordination.
7. Team members must contribute fairly and equally.
8. Conflicts must be reported promptly to the Faculty Handler / Department Coordinator / Chief Coordinator
9. Students must meet regularly with their Mentor and Faculty Handler of the class weekly.
10. Meeting notes or progress reports must be maintained in terms of Mentor / Reviewer Diary
11. Follow official timelines for proposals, drafts, final reports, and presentations.
12. Maintain the documentation phase by phase.

13. Projects must follow ethical guidelines (no falsified data, respect IP, consent/privacy for surveys).
14. Follow the formatting guidelines (APA, IEEE, etc.) for all submissions.
15. Submit all required documents: proposal, report, logbook, and presentation materials.
16. Deliver a final presentation and be prepared for questions.
17. Demonstration of project with Coverage, Completeness and Convergence is mandatory
18. Maintain professional conduct with Mentor, Faculty Handler, Teammates, and evaluators.
19. Use institutional resources responsibly and uphold academic integrity.
20. Papers must be published in Scopus-Indexed IEEE International Conferences / reputed Journals in both semesters.

DSA Components

Component	Mentor	Panel	Faculty Handler	Final marks
Review–I	25	25	–	10
Review–II	25	25	–	10
Review–III	25	25	–	10
Report, publication and Faculty Handler assessment	10	–	10	20
Mentor — Continuous supervision	20	–	–	20
Continuous Internal Evaluation				70
Final Review — Semester-End Viva Voce Examination				30
Total				100

The Four Bands on Evaluation

Every criterion in this document carries four descriptors, one per band. Where a performance falls between two descriptors, the lower band applies unless the evaluator records a reason on the sheet.

Band	Share of criterion marks	Meaning
Exemplary	85–100%	Exceeds the standard expected at this stage; independent, rigorous and defensible under questioning.
Proficient	70–84%	Meets the standard expected at this stage in full. This is the target, not a disappointment.
Developing	50–69%	Partially meets the standard; identifiable gaps that can be closed before the next review.
Inadequate	Below 50%	Does not meet the standard; the component must be redone.

Converting a Band into a Mark

Every criterion in this document is worth either 5 marks (all review criteria and the supporting heads) or 10 marks (the three Final Review criteria). The table below gives the whole-number mark for each band, so that no evaluator has to compute a percentage or decide how to round.

Criterion maximum	Exemplary	Proficient	Developing	Inadequate
5 — all review criteria, report and publication, Faculty Handler, Mentor supervision	5	4	3	2, 1 or 0
10 — Final Review: presentation, demonstration, viva voce	10 or 9	8 or 7	6 or 5	4 and below

DESIGN PROJECT – 1 (5CS1990) • REVIEW–I

Indicative timing: week 5. Focus: problem, literature, objectives, methodology and plan. The approved Semester V, Implementation scope is fixed at this review.

Criterion	Mentor 25	Panel 25	Exemplary	Proficient	Developing	Inadequate
1. Problem identification, need and scope	5	5	Problem precise, bounded and evidenced by data or stakeholder contact; scope states its exclusions; the need is demonstrated, not asserted.	Problem clear and relevant; scope defined; need supported by some evidence.	Problem stated but broad; scope loose at the edges; need asserted rather than evidenced.	No identifiable problem, or a restatement of a source; no scope set; no need established.
2. Literature review and gap analysis	5	5	Minimum 12 sources, mostly from the last 5 years, synthesised into a comparison matrix with critical commentary; the gap is defended against alternative readings.	8–12 relevant sources summarised; comparison matrix present; gap identified and traceable.	5–8 sources summarised but not synthesised; matrix incomplete; gap asserted rather than derived.	Fewer than 5 sources, or sources irrelevant or evidently unread; no matrix; no gap.
3. Objectives and requirements	5	5	Objectives measurable and time-bound; functional and non-functional requirements testable and traceable to objectives.	Objectives specific; requirements documented and mostly testable.	Objectives general; requirements listed but not testable or not traceable to objectives.	Objectives are aspirational statements; no requirements prepared.
4. Proposed methodology, tools and feasibility	5	5	Methodology justified against alternatives considered; stack chosen on performance, cost and team capability; data sources secured; risk register present.	Methodology appropriate; tools identified with reasons; feasibility argued.	Methodology outlined; tools named without justification; feasibility asserted rather than examined.	No coherent methodology; no tool selection; feasibility not considered at all.
5. Presentation and technical defence	5	5	Professional delivery; every member answers questions on their own work without deferring to one spokesperson.	Clear delivery; questions answered adequately.	Delivery uneven; some questions unanswered; one member carries the session.	Disorganised; unable to defend the approach.
TOTAL	25	25				

Review score = (Mentor + Panel) ÷ 5 = _____ / 10

DESIGN PROJECT – 1 (5CS1990) • REVIEW–II

Indicative timing: week 9. Focus: architecture, module design and 50% implementation, demonstrated live.

Criterion	Mentor 25	Panel 25	Exemplary	Proficient	Developing	Inadequate
1. System architecture	5	5	Architecture, component decomposition, interfaces and data flow complete and internally consistent; integration risks identified.	Architecture and principal diagrams complete; components defined with interfaces.	Architecture drafted but incomplete or internally inconsistent.	No architecture produced.
2. Module description	5	5	Every module defined by purpose, inputs, outputs, interface contract and failure behaviour; dependencies between modules explicit.	Modules defined with purpose, inputs, outputs and interfaces.	Modules named and broadly described; interfaces or dependencies not specified.	Modules listed without description.
3. Implementation progress — target 50% of the approved scope	5	5	At or beyond 55% of the approved module list, running and demonstrable end to end.	45–54%; the core mechanism runs and can be shown.	35–44%; something runs, but not the core mechanism.	Below 35%; little or nothing runs.
4. Presentation and viva	5	5	Professional delivery; every member answers technical questions on their own work and on how it fits the whole.	Clear delivery; technical questions answered adequately.	Delivery uneven; questions answered by one member only.	Disorganised; unable to answer questions on the work.
5. Project demonstration	5	5	Live demonstration of working functionality; behaviour explained, and failure handling shown when probed.	Live demonstration of the principal path.	Demonstration partial or shown on recorded output without reason.	No demonstration, or it fails.
TOTAL	25	25				

Review score = (Mentor + Panel) ÷ 5 = _____ / 10

DESIGN PROJECT – 1 (5CS1990) • REVIEW—III

Indicative timing: week 13. Focus: completed Semester V scope, demonstration, usability, paper submitted and report.

Criterion	Mentor 25	Panel 25	Exemplary	Proficient	Developing	Inadequate
1. Implementation completeness	5	5	Approved Semester V scope fully implemented, integrated and stable; code structured and reviewed.	Approved scope substantially implemented; minor functionality outstanding.	Majority implemented, but significant functionality outstanding or unstable.	Substantially incomplete; the core scope was not delivered.
2. Demonstration	5	5	Complete working demonstration; failure modes handled and explained under questioning.	Working demonstration of the principal path.	Demonstration partial, or it breaks when taken off the prepared path.	No live demonstration, or it fails.
3. Usability and user satisfaction	5	5	Evaluated with real users; task completion and satisfaction recorded; findings fed back into the build and the change is visible.	Usability evaluated with users or a defined protocol; findings recorded.	Interface reviewed informally within the team; no user contact.	Usability not considered.
4. Submission of the paper	5	5	Submitted with proof; manuscript of a standard the venue would plausibly accept; choice of venue justified.	Submitted, with proof of submission on file.	Manuscript complete and venue nominated, but submission not yet made.	Not submitted and no manuscript or submitted to a venue outside the approved list.
5. Report quality and conformance	5	5	Complete, IEEE-conformant, references verified against originals; reproducible from the description.	Complete and largely conformant.	Complete in outline, but sections thin or formatting inconsistent.	Incomplete or not submitted.
TOTAL	25	25				

Review score = (Mentor + Panel) ÷ 5 = _____ / 10

DESIGN PROJECT – 1 (5CS1990) • FINAL REVIEW — 30 Marks

Semester-end examination. Panel of two reviewers in the school. Includes a live demonstration.

Criterion	Max	Exemplary	Proficient	Developing	Inadequate
Presentation	10	Professionally structured and timed; the argument runs from problem to result without gaps; slides support rather than replace the speaker; interruption handled calmly.	Clear, well-organised presentation covering problem, approach, implementation and results within time.	Presentation covers the work but is unbalanced, over or under time, or reads from slides.	Disorganised or incomplete; the panel cannot follow what was done.
Demonstration	10	Complete working demonstration of the approved Semester V scope under realistic input; edge cases and failure handling shown when probed.	Working demonstration of the principal path, running live.	Demonstration partial, restricted to a rehearsed path, or shown on recorded output.	No working demonstration, or it fails.
Viva voce	10	Every member defends their own contribution and answers on parts of the system they did not build; design decisions justified against alternatives; limitations acknowledged honestly.	Members answer technical questions on their own work and justify the approach taken.	Some members answer; others defer to a spokesperson; answers stay at the level of what was done rather than why.	Unable to explain own submitted work.
TOTAL	30				

DESIGN PROJECT – 2 (6CS1991) • REVIEW—I

Indicative timing: week 4. Focus: consolidation of Semester V, extended scope and design, implementation plan and early build.

Criterion	Mentor 25	Panel 25	Exemplary	Proficient	Developing	Inadequate
1. Consolidation of Semester V work and closure of feedback	5	5	Every point from the Semester V final review closed and evidenced; the team can explain what changed and why.	Feedback largely addressed; carried-forward work is stable.	Some feedback addressed; carried-forward work runs but is essentially unchanged.	No advance on the Semester V position.
2. Refined scope, objectives and extended design	5	5	Semester VI scope is a defensible extension, not a repetition; architecture updated with interfaces and data contracts finalised.	Scope and objectives extended and documented; design updated.	Scope extended only marginally; design updated in outline only.	Scope restated from Semester V; design unchanged.
3. Implementation plan and early implementation evidence	5	5	Module-wise plan with milestones, owners and risk mitigations; extended modules already running with commits from every member.	Plan with milestones; environment operational and some new code running.	Timeline only; code written but not yet running.	No credible plan; nothing new built since Semester V.
4. Presentation and technical defence	5	5	Confident, evidence-based presentation; every member accountable for their own part.	Clear presentation; questions answered.	Presentation adequate, but questions answered by one member only.	Unprepared; unable to defend the extended scope.
5. Project demonstration	5	5	Carried-forward system demonstrated running, with the new work clearly distinguished from the Semester V build.	Carried-forward system demonstrated running.	Demonstration partial or shown on recorded output.	No demonstration, or it fails.
TOTAL	25	25				

Review score = (Mentor + Panel) ÷ 5 = _____ / 10

DESIGN PROJECT – 2 (6CS1991) • REVIEW–II

Indicative timing: week 9. Focus: integration design, module development and 70% implementation, demonstrated live.

Criterion	Mentor 25	Panel 25	Exemplary	Proficient	Developing	Inadequate
1. System architecture and integration design	5	5	Integration design complete and consistent; interfaces, data contracts and failure paths specified across all modules.	Architecture updated; interfaces defined for the modules being integrated.	Architecture updated in outline; integration approach not specified.	No integration design.
2. Module development and technical quality	5	5	Modules built, reviewed and version-controlled; error handling and security considerations addressed.	Modules built and functional; work under version control.	Modules run individually; code unstructured or unreviewed.	Little working code; no version control discipline.
3. Implementation progress — target 70% of the extended scope	5	5	At or beyond 77% of the approved module list, integrated and demonstrable.	63–76%; core modules integrated and demonstrable.	49–62%; partial functionality running.	Below 49%; little running code.
4. Presentation and viva	5	5	Professional delivery; every member answers technical questions on their own work and on the system as a whole.	Clear delivery; technical questions answered adequately.	Delivery uneven; questions answered by one member only.	Disorganised; unable to answer questions on the work.
5. Project demonstration	5	5	Live demonstration of integrated functionality, with failure handling shown when probed.	Live demonstration of the principal path.	Demonstration partial, or of modules in isolation rather than integrated.	No demonstration, or it fails.
TOTAL	25	25				

Review score = (Mentor + Panel) ÷ 5 = _____ / 10

DESIGN PROJECT – 2 (6CS1991) • REVIEW—III

Indicative timing: week 13. Focus: 100% implementation, validation against baseline, paper submitted and report.

Criterion	Mentor 25	Panel 25	Exemplary	Proficient	Developing	Inadequate
1. Implementation completeness — 100% of the extended scope	5	5	Full approved scope implemented, integrated and stable; deviations from the design documented and justified.	Nearly complete; minor functionality outstanding.	Majority complete; significant functionality outstanding.	Substantially incomplete.
2. Demonstration	5	5	Complete demonstration under realistic conditions, including edge cases and failure handling.	Working demonstration of the full principal path.	Demonstration partial, or restricted to a rehearsed path.	Demonstration incomplete or failed.
3. Testing, validation and results against baseline	5	5	Coverage evidence; validation traced to requirements; results compared against baseline and literature and interpreted critically including anomalies.	Comprehensive test cases executed; results presented with appropriate metrics.	Testing performed; results presented without a baseline or without interpretation.	No systematic testing.
4. Submission of the paper	5	5	Submitted to a Scopus-indexed venue with proof, reporting the actual results; acceptance obtained, or the manuscript is plainly of acceptable standard.	Submitted, with proof of submission on file.	Manuscript complete and venue nominated, but submission not yet made.	Not submitted, or submitted to a venue outside the approved list.
5. Report quality and conformance	5	5	Complete, IEEE-conformant, references verified; a third party could reproduce the work from it.	Complete and largely conformant.	Complete in outline; sections thin or formatting inconsistent.	Incomplete or not submitted.
TOTAL	25	25				

Review score = (Mentor + Panel) ÷ 5 = _____ / 10

DESIGN PROJECT – 2 (6CS1991) · FINAL REVIEW — 30 Marks

Semester-end examination. Panel of two members in the school. Includes a full working demonstration under realistic conditions.

Criterion	Max	Exemplary	Proficient	Developing	Inadequate
Presentation	10	Professionally structured and timed; the argument runs from problem to result without gaps; slides support rather than replace the speaker; interruption handled calmly.	Clear, well-organised presentation covering problem, approach, implementation and results within time.	Presentation covers the work but is unbalanced, over or under time, or reads from slides.	Disorganised or incomplete; the panel cannot follow what was done.
Demonstration	10	Complete working demonstration of the full system under realistic conditions; edge cases, load and failure handling shown when probed.	Working demonstration of the principal path, running live.	Demonstration partial, restricted to a rehearsed path, or shown on recorded output.	No working demonstration, or it fails.
Viva voce	10	Every member defends their own contribution and answers on parts of the system they did not build; design decisions justified against alternatives; limitations acknowledged honestly.	Members answer technical questions on their own work and justify the approach taken.	Some members answer; others defer to a spokesperson; answers stay at the level of what was done rather than why.	Unable to explain own submitted work.
TOTAL	30				

REPORT AND PUBLICATION — 10 Marks (Both Semesters)

Assessed by the Mentor and countersigned by the Chief Coordinator.

Criterion	Max	Exemplary	Proficient	Developing	Inadequate
Report — structure, technical content and referencing	5	Complete and IEEE-conformant; analysis goes beyond description and design decisions are justified; all references verified against the original source.	Complete and largely conformant; sound description with some analysis; references accurate.	Complete in outline; descriptive with little analysis; references present but some uncited or inconsistently formatted.	Incomplete or not submitted; a diary of activities rather than a technical report; references unverified or fabricated.
Publication — manuscript quality and evidence of submission	5	Submitted to a venue on the approved list with proof, written to that venue's standard; accepted or published within the semester where the calendar allowed.	Submitted to a venue on the approved list, with proof of submission on file.	Manuscript complete and venue nominated from the approved list, but submission not yet made.	No manuscript, no submission, or submitted to a venue outside the approved list.
TOTAL	10				

FACULTY HANDLER ASSESSMENT — 10 Marks (Both Semesters)

Assessed by the Faculty Handler, independently of the Mentor, so that records and deadlines are verified by someone other than the person who signs the records. This head covers process and discipline only; the substance of the work is assessed in the reviews.

Criterion	Max	Exemplary	Proficient	Developing	Inadequate
Adherence to deadlines and submission discipline	5	Every submission on or before the notified date, in the prescribed format, without reminder.	All submissions made; at most one late, with a recorded reason.	Two or more late submissions, or repeated reminders required.	Deadlines routinely missed; submissions incomplete or in the wrong format.
Completeness and currency of project records	5	Review diary complete with 12 or more entries signed at each meeting; all forms, similarity report and AI-usage declaration filed on time and in order.	Diary complete with minor gaps; all required documents filed.	8–11 diary entries, or some signed retrospectively; documents filed only after reminder.	Fewer than 8 entries, or signed in bulk at the end; required documents missing.
TOTAL	10				

MENTOR — CONTINUOUS SUPERVISION — 20 Marks (Both Semesters)

Assessed once at the end of the semester, on the evidence of the diary, the meeting record and the work seen throughout. These four criteria are what the Mentor is positioned to judge across a whole semester and a single review panel is not.

Criterion	Max	Exemplary	Proficient	Developing	Inadequate
Idea, innovation and novelty	5	The problem and approach show genuine originality; the team can say precisely what is theirs and defend why it is worth doing.	A sound, relevant idea competently developed; some original element beyond routine application.	A workable but derivative idea; largely an application of a known solution to a known problem.	No discernible idea of the team's own; a reproduction of an existing tutorial or project.
Coverage, completeness and convergence	5	Every objective is covered by delivered work; no orphan requirements; the work converges on the problem rather than drifting from it.	Most objectives covered by delivered work; scope held steady through the semester.	Gaps between objectives and delivered work; scope drifted and was not formally rescoped.	The work does not address the objectives set; scope abandoned without record.
Contribution to the report and publication	5	Sustained, evidenced contribution by every member to writing and revision; drafts improved materially across iterations.	All members contributed to the report and the paper; drafts revised on feedback.	Contribution uneven; writing concentrated in one or two members.	Writing not attributable or produced at the last moment without revision.
Academic integrity and plagiarism compliance	5	Similarity within threshold at first submission; AI use fully declared; every member can explain any part of the submitted work and code on demand.	Similarity within threshold; declarations complete and accurate.	Similarity brought within threshold only after revision; declarations filed late or amended after questioning.	Similarity above threshold, declarations missing or inaccurate, or submitted work the student cannot explain.
TOTAL	20				